Graph Neural Network-Based Anticancer Drug Response Prediction Using Molecular Graphs and Cell-Line Information

Abstract

Anticancer drug response prediction is an important task in computational oncology because experimental drug screening is expensive and time-consuming. Machine learning and deep learning methods can help estimate drug sensitivity by learning relationships between drug properties and cancer cell-line characteristics. In this project, graph neural network-based models were implemented for anticancer drug response prediction using the GDSC2 dataset. Drug molecules were represented as molecular graphs created from SMILES strings, where atoms were treated as nodes and chemical bonds as edges. The prediction target was LN_IC50, a continuous measure of drug sensitivity.

Two main models were developed and compared. The first model was a GNN-only baseline that used only molecular graph information. The second model was a hybrid GNN model that combined molecular graph embeddings with cancer cell-line embeddings. In the warm-start experiment, the hybrid GNN achieved the best performance, with a test RMSE of 1.3950, test MAE of 1.0619, and Pearson correlation of 0.8758. This outperformed the GNN-only baseline, which achieved a test RMSE of 1.5481 and Pearson correlation of 0.8429. A cold-drug experiment was also performed to evaluate generalization to unseen drugs. In this setting, performance decreased substantially, with a test Pearson correlation of 0.2431. These results show that combining drug molecular graphs with cell-line information improves prediction in warm-start settings, while generalization to unseen drugs remains a major challenge.

Keywords: Anticancer drug response prediction; graph neural networks; GDSC2; molecular graphs; SMILES; IC50 prediction; deep learning; cancer cell lines; cold-drug split; hybrid GNN

Introduction

Cancer treatment response can vary strongly between patients and cancer types. A drug that is effective for one cancer cell line or tumor type may have limited effect on another. Because of this variability, predicting anticancer drug response is an important problem in computational oncology. Accurate prediction models can help identify promising drug candidates, support drug screening, and reduce the cost and time required for experimental testing.

Traditional anticancer drug discovery and response evaluation often rely on laboratory screening. Although experimental drug sensitivity testing provides direct biological evidence, it is expensive, time-consuming, and difficult to scale to large numbers of drugs and cancer models. Because of these limitations, computational methods have increasingly been used to predict anticancer drug activity and drug response. Earlier machine learning approaches used chemical descriptors, molecular fingerprints,

gene expression, mutation features, and similarity-based methods to estimate drug sensitivity or classify anticancer compounds [2]–[7].

More recently, deep learning methods have become popular for drug response prediction because they can learn complex nonlinear relationships from high-dimensional biological and chemical data. Studies such as DeepSynergy and other drug synergy models showed that combining drug-related information with cancer cell-line features can improve prediction of drug effects [9], [10]. Other recent models, including MOViDA, DrugGene, TINDL, and TransCDR, further showed the importance of multimodal learning, biological interpretability, transfer learning, and generalization to unseen drugs or cell lines [12]–[15].

Graph neural networks are especially suitable for drug-related prediction tasks because drug molecules naturally have graph structure. In a molecular graph, atoms can be represented as nodes and chemical bonds as edges. Unlike traditional descriptor-based methods, GNNs can learn representations directly from molecular structure. This makes them useful for modeling drug properties and predicting drug response from SMILES-derived molecular graphs.

The aim of this project is to implement and evaluate graph neural network-based models for anticancer drug response prediction using the GDSC2 dataset. The first model is a GNN-only baseline that uses only molecular graph information from drugs. The second model is a hybrid GNN model that combines drug molecular graph embeddings with cancer cell-line embeddings. In addition to a standard warm-start experiment, a cold-drug split is used to test whether the model can generalize to drugs that were not seen during training.

The main contribution of this project is the practical comparison between a GNN-only model and a hybrid GNN model for LN_IC50 prediction. The results show that adding cell-line information improves performance in the warm-start setting. However, the cold-drug experiment shows that generalization to unseen drugs remains difficult, which is consistent with challenges reported in previous drug response prediction studies.

Experimental Setup

The practical implementation was performed using Python. The main libraries used were PyTorch, PyTorch Geometric, RDKit, pandas, scikit-learn, NumPy, and matplotlib. PyTorch was used for building and training the neural network models, while PyTorch Geometric was used to represent and process molecular graphs. RDKit was used to convert drug SMILES strings into molecular graph structures.

The experiments were performed on a CPU-based setup using Python 3.9.13 and PyTorch 2.8.0+cpu. Because GPU acceleration was not available, a CPU-friendly subset of the processed GDSC2 dataset was used for training. From the full processed dataset, 30,000 records were sampled for the warm-start experiment. This sample was divided into 24,000 training records, 3,000 validation records, and 3,000 test records

For the cold-drug experiment, the split was created by drug identifier rather than by individual rows. The training set contained 195 drugs, the validation set contained 24 drugs, and the test set contained 25 drugs. There was no drug overlap between the training, validation, and test sets. This setup was used to evaluate whether the model could generalize to drugs that were completely unseen during training.

Both the GNN-only and hybrid GNN models were trained for 50 epochs. The batch size was 64, the learning rate was 0.001, and the Adam optimizer was used with weight decay of 1e-5. Mean squared error was used as the loss function because the task was formulated as regression. During training, validation performance was monitored after each epoch, and the model with the lowest validation RMSE was saved as the best model.

Model performance was evaluated using RMSE, MAE, and Pearson correlation. RMSE and MAE measure prediction error, while Pearson correlation measures how strongly predicted LN_IC50 values follow the same trend as the actual values. Lower RMSE and MAE indicate better prediction accuracy, while higher Pearson correlation indicates stronger agreement between predicted and actual drug response values.

Related Work

Early work on anticancer drug response prediction was based on experimental drug sensitivity testing rather than modern machine learning. One early methodological study investigated whether anticancer drug effects observed in tissue culture could be used to predict drug response in animal tumor models and human tumor samples [1]. Although this type of approach showed that in vitro response can provide useful information about treatment effect, it also demonstrated the practical limitation of relying only on experimental screening. This limitation motivated later computational approaches that aim to reduce the cost and time required for identifying promising anticancer compounds.

Several studies used classical machine learning and similarity-based methods for anticancer activity prediction. CDRUG used compound fingerprints and a hybrid similarity score to predict whether a chemical compound has anticancer activity based on active and inactive compounds from the NCI-60 Developmental Therapeutics Program dataset [2]. A related hybrid anticancer molecule prediction study also used NCI-60 screened compounds and combined support vector machine predictions with a potency score, reporting strong performance and implementing the method in the CancerIN web server [3]. These studies show that chemical structure information alone can be useful for early-stage anticancer compound screening, especially when compounds can be represented using SMILES, fingerprints, or molecular similarity scores.

Other classical machine learning studies combined drug chemical information with cancer cell-line features. Lind and Anderson used random forest models to predict drug activity against cancer cells based on a limited set of oncogene mutation features and chemical descriptors [4]. Their work is important because it showed that useful prediction can be achieved even with relatively minimal genomic input, as long as drug structural descriptors are included. Similarly, gene-expression-based drug response prediction studies used GDSC data, ElasticNet regression, IC50/AUC response values, and predictor-gene selection to identify genes associated with anticancer drug sensitivity [5]. These studies provide useful baselines for the field, but they depend on manually selected descriptors, fingerprints, or feature selection steps rather than learning molecular structure directly.

Matrix factorization methods have also been used for anticancer drug response prediction in cell lines. WGRMF represented drug response as a matrix and incorporated drug similarity and cell-line similarity through graph regularization [6]. In that study, GDSC drug response was represented using log-transformed IC50 values, drugs were characterized using PubChem fingerprints, and cell lines were characterized using gene expression profiles. A related SRMF method also used drug chemical structure similarity and cell-line gene expression similarity to improve missing drug response prediction on GDSC

and CCLE datasets [7]. These approaches are relevant because they explicitly model relationships between drugs and cell lines, but they still rely on predefined similarity matrices instead of directly learning molecular graph representations.

Deep learning has become increasingly important in anticancer drug response prediction because drug response is influenced by nonlinear relationships between chemical structure and cancer biology. A review on deep learning for drug response prediction summarizes the use of architectures such as fully connected networks, CNNs, RNNs, autoencoders, variational autoencoders, and deep belief networks, and also discusses common datasets such as GDSC, CCLE, CTRP, and NCI-60 [8]. The same review highlights important limitations in the field, including data quality, interpretability, model generalization, and the difficulty of transferring cell-line predictions to patient-level outcomes. These issues are directly relevant to this project because the practical work compares warm-start performance with cold-drug generalization.

Drug synergy prediction is a related but more complex task because it tries to predict the effect of drug combinations rather than single-drug response. DeepSynergy is one of the key deep learning studies in this area. It used both drug chemical information and cancer cell-line genomic information to predict synergistic drug combinations and compared the deep learning model against methods such as gradient boosting machines, random forests, support vector machines, and elastic nets [9]. DeepSynergy achieved strong performance for novel combinations within already explored drugs and cell lines, but its performance decreased when extrapolating to unexplored drugs or cell lines. This finding is important for the present project because it supports the idea that cold-start prediction is much harder than warm-start prediction.

Further work on drug combination prediction used multi-omics and hybrid machine learning approaches. DeepMDS integrated cancer cell-line gene expression, anticancer drug target information, and drug response data to predict synergistic multi-drug combinations [10]. Another study developed a machine learning model for anticancer drug combinations using multiple synergy metrics, including ZIP, Loewe, Bliss, and HSA, and included experimental validation in 2D cancer cell lines and 3D tumor slice culture models [11]. These studies show that combining drug information with cancer-specific biological context can improve prediction and make models more useful for identifying candidate therapies.

Several recent studies focused on interpretability and biologically informed modeling. MOViDA introduced a multi-omics visible neural network for drug activity prediction, using Gene Ontology biological processes to structure part of the model and improve biological interpretability [12]. The model combined mutation, copy number variation, gene expression pathway activity, PubChem fingerprints, and molecular descriptors, and it was also extended to drug synergy prediction. DrugGene similarly used an interpretable deep learning structure guided by biological processes, combining cell-line features such as gene expression, mutation, and copy number variation with drug chemical fingerprints [13]. These studies show that modern drug response models are moving beyond prediction accuracy alone and are increasingly expected to provide biological explanations.

The challenge of transferring models from preclinical cell-line data to clinical patient response has also been studied. TINDL addressed this problem by training on preclinical cancer cell-line data and applying tissue-informed normalization to improve prediction of clinical drug response in TCGA patients [14]. The study also used biomarker identification and siRNA knockdown validation to support the biological relevance of selected genes. This is important because it shows that even strong models trained on cell

lines may not directly generalize to patient tumors without additional normalization or domain adaptation.

Transfer learning and multimodal fusion have recently become important for improving model generalization. TransCDR combined multiple drug representations, including SMILES strings, molecular graphs, and ECFP fingerprints, with multiple cancer cell-line omics profiles, including mutation/copy number aberration, gene expression, and DNA methylation [15]. It also used pretrained drug encoders and self-attention-based fusion instead of simple concatenation. The authors reported improvements across warm and cold-start scenarios, but also noted that cold scaffold prediction remains difficult and that improved molecular representations, including geometry-aware GNNs, may be needed. This paper is especially relevant to the current project because it directly supports the use of molecular graphs, hybrid drug-cell representations, and cold-start evaluation.

Computational docking and pharmacokinetic prediction have also been explored as additional sources of drug-related information. One study integrated large-scale docking scores into anticancer drug response prediction models to test whether protein-ligand docking features improve performance when combined with gene expression and drug descriptors [16]. The results showed only marginal improvement, suggesting that docking scores may add some mechanistic information but can overlap with chemical descriptors. Other studies used pharmacokinetic and ADMET prediction platforms to prioritize anticancer compounds based on properties such as CYP metabolism, hERG toxicity, drug-likeness, and docking behavior [17]. These works show that drug response prediction can be improved or supported by additional chemical and biological information, although not every added feature type improves performance equally.

Graph-based representations are especially relevant for anticancer drug prediction because molecules naturally have graph structure. Earlier structured-output and graph-kernel approaches used molecular graph representations for anticancer activity prediction across multiple cancer cell lines [18]. More recent deep learning approaches, such as TransCDR, include molecular graphs as one of the drug input modalities, while other studies emphasize the need for better molecular representation learning in cold-start settings [15]. This directly motivates the present project, where SMILES strings from GDSC2 drugs were converted into molecular graphs and used as input to a graph neural network.

Overall, the reviewed literature shows a clear progression from experimental sensitivity testing and classical machine learning toward deep learning, multimodal fusion, interpretability, and transfer learning. Classical methods such as random forests, SVMs, ElasticNet, QSAR, and matrix factorization provide useful baselines, while newer methods increasingly combine drug structure with cancer cell-line or omics information. The main gap repeatedly observed in the literature is generalization: models often perform well in warm-start settings but perform worse when tested on unseen drugs, unseen cell lines, or patient-level data. Therefore, this project focuses on graph neural network-based anticancer drug response prediction using GDSC2, compares a GNN-only model with a hybrid GNN model, and includes a cold-drug experiment to test unseen-drug generalization.

Dataset

This project used the GDSC2 drug response dataset for anticancer drug response prediction. The original dataset contained 242,036 drug response records. Each record included information about a cancer cell line, a tested drug, and response values such as LN_IC50, AUC, and Z-score. In this project, LN_IC50 was

selected as the main prediction target because it represents drug sensitivity in a continuous form and is commonly used in anticancer drug response prediction studies.

Since the proposed model is based on graph neural networks, molecular structure information was required for each drug. The original GDSC2 file contained drug names and drug identifiers, but it did not directly include SMILES strings. Therefore, an additional preprocessing step was performed to retrieve SMILES representations for the drugs. Out of 295 unique drugs, SMILES strings were successfully collected for 244 drugs. After merging the GDSC2 response data with the SMILES mapping, the final usable dataset contained 204,145 records, 244 drugs, and 963 cancer cell lines.

For the first experiment, a warm-start split was used. From the full processed dataset, a CPU-friendly sample of 30,000 records was selected. This sample was divided into 24,000 training records, 3,000 validation records, and 3,000 test records. In this setting, drugs and cell lines may appear across training and test sets, but the individual drug-cell line response pairs are separated.

In addition to the warm-start experiment, a cold-drug split was created to evaluate generalization to unseen drugs. In this split, the dataset was divided based on drug identifiers rather than individual rows. The training set contained 195 drugs, the validation set contained 24 drugs, and the test set contained 25 drugs. There was no overlap between drugs in the training, validation, and test sets. This made the cold-drug experiment more difficult because the model had to predict responses for drugs that were not observed during training.

Methodology

The main goal of the practical part was to predict anticancer drug response using graph neural network-based models. Two main models were implemented and compared: a GNN-only baseline and a hybrid GNN model.

SMILES to molecular graph conversion

Each drug was represented using its SMILES string. The SMILES string was converted into a molecular graph using RDKit. In the molecular graph, atoms were treated as nodes and chemical bonds were treated as edges. Since molecular bonds are undirected, each bond was added in both directions in the graph representation.

For each atom, five basic atom-level features were used:

- atomic number
- atom degree
- formal charge
- total number of hydrogens
- aromaticity indicator

These node features were used as input to the graph neural network. Each graph was then stored in PyTorch Geometric format, together with the target LN_IC50 value and the corresponding cancer cell-line index.

GNN-only baseline model

The first model was a GNN-only baseline. This model used only the molecular graph of the drug as input and did not include any cell-line information. The purpose of this model was to test how much predictive information could be learned from the drug structure alone.

The model used three graph convolutional layers. After graph convolution, global mean pooling was applied to convert node-level embedding's into one graph-level drug embedding. This drug embedding was then passed through fully connected prediction layers to estimate LN_IC50.

Hybrid GNN model

The second model was a hybrid GNN model. This model combined molecular graph information with cancer cell-line information. The drug branch used the same GCN-based molecular graph encoder as the GNN-only model. In addition, each cancer cell line was represented using an embedding layer. The drug embedding and cell-line embedding were then concatenated and passed into prediction layers to estimate LN_IC50.

This hybrid design was used because drug response depends not only on the drug structure, but also on the biological context of the cancer cell line. Therefore, adding cell-line information was expected to improve prediction performance compared with using the drug graph alone.

Cold-drug evaluation

To test model generalization, an additional cold-drug experiment was performed. In this setting, the hybrid GNN model was trained on one group of drugs and tested on a separate group of drugs that were never seen during training. This experiment is more challenging than the warm-start setting because the model cannot rely on having already seen the same drugs during training.

Evaluation metrics

The models were evaluated using three regression metrics: RMSE, MAE, and Pearson correlation.

RMSE measures the average prediction error with stronger penalty for larger errors. MAE measures the average absolute difference between predicted and actual LN_IC50 values. Pearson correlation measures how strongly the predicted and actual values follow the same trend. Lower RMSE and MAE values indicate better prediction accuracy, while a higher Pearson correlation indicates better agreement between predicted and actual drug response patterns.

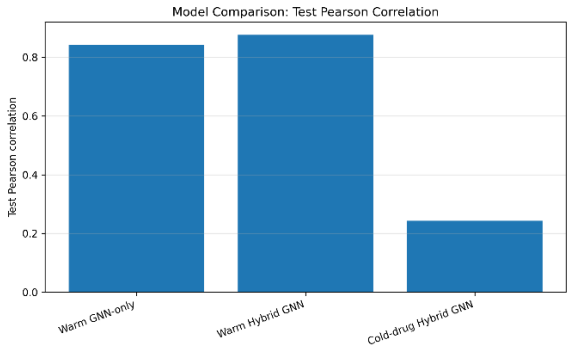
Results

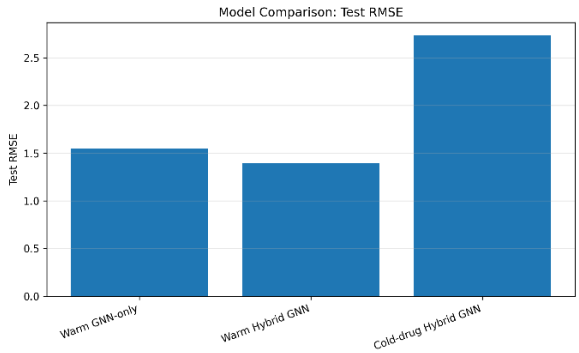
Model	RMSE	MAE	Pearson
Warm GNN-only	1.5481	1.1755	0.8429
Warm Hybrid GNN	1.3950	1.0619	0.8758
Cold-drug Hybrid GNN	2.7325	2.1175	0.2431

Table 1. Model comparison results

The warm-start hybrid GNN achieved the best overall performance, with a test RMSE of 1.3950, test MAE of 1.0619, and Pearson correlation of 0.8758. Compared with the GNN-only baseline, the hybrid model reduced RMSE from 1.5481 to 1.3950 and improved Pearson correlation from 0.8429 to 0.8758. This

shows that adding cell-line information improved drug response prediction compared with using only molecular graph information.





The GNN-only baseline also achieved strong performance, with a Pearson correlation of 0.8429. This suggests that molecular graph structure alone contains useful information for predicting anticancer drug response. However, the stronger performance of the hybrid model shows that drug structure is not sufficient by itself and that cancer cell-line context provides additional useful information.

The cold-drug hybrid experiment produced lower performance, with a test RMSE of 2.7325 and Pearson correlation of 0.2431. This result was expected because the test drugs in the cold-drug split were completely unseen during training. Therefore, the model had to generalize to new molecular structures without prior response examples for those drugs. The drop in performance shows that cold-drug prediction is significantly more difficult than warm-start prediction.

Overall, the results show that the proposed hybrid GNN model is effective in the warm-start setting and improves over the GNN-only baseline. At the same time, the cold-drug experiment shows that generalization to unseen drugs remains a major challenge and should be considered an important direction for future work.

Discussion

The results of this project show that graph neural networks can be used effectively for anticancer drug response prediction when molecular structure information is available. The GNN-only baseline achieved a test Pearson correlation of 0.8429, which indicates that molecular graph structure alone contains useful predictive information. This is expected because the chemical structure of a drug is directly related to its biological activity and interaction with cancer-related targets.

However, the hybrid GNN model achieved better performance than the GNN-only baseline in the warm-start setting. The hybrid model reduced the test RMSE from 1.5481 to 1.3950 and improved the Pearson correlation from 0.8429 to 0.8758. This suggests that adding cancer cell-line information improves prediction performance. Drug response does not depend only on the drug molecule itself, but also on the biological context of the tested cancer cell line. Therefore, the cell-line embedding helped the model learn differences between cancer cell lines and improved the prediction of LN_IC50 values.

The warm-start results are encouraging because they show that the model can learn meaningful relationships between drug molecular structure, cancer cell-line identity, and drug response. In this

setting, the same drugs and cell lines may appear across training and test data, although the exact drug-cell line response records are separated. This makes the task realistic for predicting missing responses among already known drugs and cell lines.

The cold-drug experiment was more challenging. In this setting, the test drugs were completely unseen during training. The cold-drug hybrid model achieved a test RMSE of 2.7325, test MAE of 2.1175, and Pearson correlation of 0.2431. This performance is much lower than the warm-start result. However, this does not mean that the model failed. Instead, it shows that generalizing to completely new drugs is significantly harder than predicting responses for drugs already represented in the training data. Additionally, it is worth mentioning that during training, the best validation Pearson reached 0.5080, but final test Pearson dropped to 0.2431. This suggests that cold-drug performance is unstable and strongly depends on which unseen drugs appear in the test set.

This result is important because it supports a common limitation in anticancer drug response prediction research. Many models perform well when training and test sets share drugs or cell lines, but performance decreases when the model is tested on unseen drugs. This suggests that stronger molecular representation learning may be needed for cold-drug prediction. Possible improvements include using more advanced GNN architectures such as GAT or GIN, adding pre-trained molecular embedding's, using transformer-based SMILES encoders, or combining molecular graphs with additional biological features such as gene expression, mutation, or copy number variation.

Another limitation of the current hybrid model is that cell lines were represented only using an embedding layer. This allows the model to learn a representation for each cell line, but it does not directly use biological features of the cell line. A more advanced version of this project could replace the simple cell-line embedding with gene expression or multi-omics data. For example, gene expression could be processed using a 1D-CNN or another encoder and then fused with the GNN drug embedding. This would make the model more biologically informative and closer to recent hybrid deep learning approaches in the literature.

Conclusion

This project implemented graph neural network-based models for anticancer drug response prediction using the GDSC2 dataset. Drug molecules were represented as molecular graphs created from SMILES strings, where atoms were treated as nodes and chemical bonds as edges. The prediction target was LN_IC50, which represents drug sensitivity as a continuous value.

Two main models were developed and compared. The first model was a GNN-only baseline that used only drug molecular graph information. The second model was a hybrid GNN model that combined molecular graph embeddings with cancer cell-line embeddings. In the warm-start experiment, the hybrid GNN achieved the best performance, with a test RMSE of 1.3950, test MAE of 1.0619, and Pearson correlation of 0.8758. This result shows that adding cell-line information improved prediction compared with using molecular graph information alone.

A cold-drug experiment was also performed to test generalization to unseen drugs. In this setting, the hybrid model achieved lower performance, with a test Pearson correlation of 0.2431. This result shows that predicting responses for completely unseen drugs is much more difficult than warm-start prediction. Therefore, the project confirms that cold-start drug response prediction remains an important challenge.

The main contribution of this project is the implementation and comparison of GNN-based and hybrid GNN-based approaches for anticancer drug response prediction. The results show that molecular graphs are useful for learning drug representations and that adding cell-line information improves prediction performance in the warm-start setting. Future work could improve the model by using gene expression or other omics data instead of simple cell-line embeddings, testing more advanced GNN architectures such as GAT or GIN, adding attention-based fusion, and using pretrained molecular representations to improve cold-drug generalization.

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